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International perspectives on artificial intelligence in higher education: An explorative study of students' intention to use ChatGPT across the Nordic countries and the USA

Montathar Faraon[1] · Kari Rönkkö[1] · Marcelo Milrad[2] · Eric Tsui[3]

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Abstract
This study explored factors influencing ChatGPT adoption among higher education students in five Nordic countries (Sweden, Finland, Denmark, Norway, and Iceland) and the USA. The unified theory of acceptance and use of technology 2 (UTAUT2) framework was employed and extended to incorporate personal innovativeness. Data was collected from 586 students recruited through Prolific and analyzed using partial least squares structural equation modeling (PLS-SEM). The findings revealed varying patterns of relationships between different factors and behavioral intention in each region. In the Nordic countries, performance expectancy, hedonic motivation, and habit demonstrated positive relationships with behavioral intention. In the USA, the results revealed positive relationships between behavioral intention and performance expectancy, social influence, habit, and personal innovativeness. Performance expectancy emerged as the strongest predictor of behavioral intention in both regions. In both the Nordic countries and the USA, habit and behavioral intention emerged as the only predictors of ChatGPT use behavior. Behavioral intention demonstrated a marginally stronger influence on use behavior in both regions. These findings offer insights for educators and policymakers regarding AI integration in academic settings by highlighting common drivers and differences in AI adoption patterns.

Keywords Artificial intelligence · Higher education · Students · Unified theory · ChatGPT · UTAUT2

1 Introduction

Artificial intelligence (AI) has become a transformative technology that reshapes how people learn, work, and interact in society (Dwivedi et al., 2023; Sætra, 2023). At its core, AI refers to computer systems capable of mimicking human

Extended author information available on the last page of the article

cognitive abilities such as understanding, reasoning, communicating, and learning (Rai et al., 2019). One of the evolving subfields of artificial intelligence is generative AI, which focuses on producing new content, such as text, images, audio, and video, by analyzing and learning from existing data patterns (Castelli & Manzoni, 2022). An example of generative AI is the development of chatbots, broadly defined as "automated conversational agents that use natural language processing and machine learning algorithms to interact with users in a human-like manner" (Kooli, 2023, p. 2).

One of the chatbots that has received increased attention is ChatGPT, a large language model (LLM) developed by OpenAI. It was released in November 2022 (OpenAI, 2022) and attracted 100 million users within two months of its launch (Milmo, 2023). It has captured the attention of educators and researchers in higher education because of its ability to transform learning practices (Aljanabi et al., 2023; Lai et al., 2023). ChatGPT has been adopted as a pedagogical agent to stimulate students' curiosity and strengthen their ability to formulate questions (Abdelghani et al., 2023). It has also been used to provide personalized assistance to students by summarizing papers, generating practice materials, and offering feedback on writing (Halaweh, 2023). However, the increasing dependency on AI tools, such as ChatGPT, among students raises concerns about potential negative consequences, such as increased laziness and reduced critical thinking skills (Gonsalves, 2024; Zhang et al., 2024).

Although studies have explored the adoption and use behavior of ChatGPT using UTAUT2 (Biloš & Budimir, 2024; Habibi et al., 2023), further empirical research is valuable for two reasons. First, incorporating additional factors, such as personal innovativeness, can provide insights into the drivers of students' adoption of ChatGPT. Second, most existing studies on higher education students' behavioral intention and use of ChatGPT are limited to single-country contexts and lack intercontinental comparisons (Habibi et al., 2023; Polyportis & Pahos, 2024; Strzelecki, 2023). This is a noteworthy limitation as regional differences may influence students' perceptions, attitudes, and use of technology (Tarhini et al., 2015).

Based on previous considerations, this study aims to address the mentioned gaps by exploring the adoption of ChatGPT among higher education students in the Nordic countries (Sweden, Finland, Denmark, Norway, and Iceland) and the United States of America, henceforth the USA. These regions were chosen for their unique and contrasting educational contexts, technological landscapes, and cultural attitudes towards technology. The Nordic countries are known for their progressive educational systems, high levels of digital literacy, and strong emphasis on student autonomy and critical thinking (Klette, 2018). In contrast, the USA has a competitive educational environment, characterized by a strong focus on individual achievement, standardized testing, and often rivalry for admission to top universities and academic scholarships (Lorbeer, 2020). This competitiveness is also reflected in technology access and usage variations among students. By comparing these regions, this study aims to provide a better understanding of the factors influencing ChatGPT adoption across different educational settings and regions. The UTAUT2 framework, extended to include the construct of personal innovativeness, was used to address the

following questions: (1) Which factors influence students' behavioral intention and actual use of ChatGPT? (2) How do these factors differ between the two regions? These were addressed using a quantitative methodological approach.

This study contributes to the existing literature on AI adoption in higher education by providing empirical insights into regional differences in ChatGPT adoption patterns between the Nordic countries and the USA. It extends the UTAUT2 framework by incorporating personal innovativeness and demonstrating how individual traits interact with technological and social factors in AI adoption. Additionally, it contributes to theory development by validating the predictive power of the extended UTAUT2 model in explaining ChatGPT adoption across different higher education contexts. Finally, the study provides a deeper understanding of ChatGPT use in higher education by identifying the key drivers and regional variations in student adoption patterns.

2 Literature review

2.1 Unified theory of acceptance and use of technology 2 (UTAUT2)

Over the past three decades, several frameworks have been proposed and modified to predict the acceptance and use of technological innovations. Initially, the technology acceptance model (TAM) was used to understand technology adoption (Davis, 1989). The TAM model continues to be adapted and extended to study emerging technologies in education, as demonstrated by recent work examining metaverse-based learning platforms (Al-Adwan et al., 2023). The TAM model gradually evolved into the UTAUT framework, which included four exogenous constructs (i.e., performance expectancy, effort expectancy, social influence, and facilitating conditions), two endogenous constructs (i.e., behavioral intention and use behavior), and four moderators (i.e., age, gender, experience, and voluntariness of use) (Venkatesh et al., 2003). The framework was later expanded to UTAUT2 and added three new constructs, namely hedonic motivation, price value, and habit (Venkatesh et al., 2012).

Moreover, Venkatesh et al. (2016) suggested that the UTAUT2 framework is particularly well suited to examine technology acceptance during the early stages of adoption, as it can account for 74 percent of the variance in behavioral intention and 54 percent in actual use. The fact that ChatGPT was launched in November 2022 makes the UTAUT2 framework a suitable choice to investigate its adoption. In addition, it has been proposed that including additional factors in the UTAUT2 framework could enhance its explanatory power (Venkatesh et al., 2016). Therefore, this study extends the UTAUT2 framework by adding the construct of personal innovativeness, which has been demonstrated to be a predictor of behavioral intention in information technology contexts (Dhiman et al., 2020; Koivisto et al., 2016; Strzelecki, 2023) (see Fig. 1).

Studies have used the UTAUT2 framework to examine students' behavioral intention and use of technology in higher education (Chandradasa & Galhena, 2022; Garrido-Gutiérrez et al., 2023; Noble et al., 2022; Zacharis & Nikolopoulou, 2022). One study in Greece examined students' adoption of the platforms

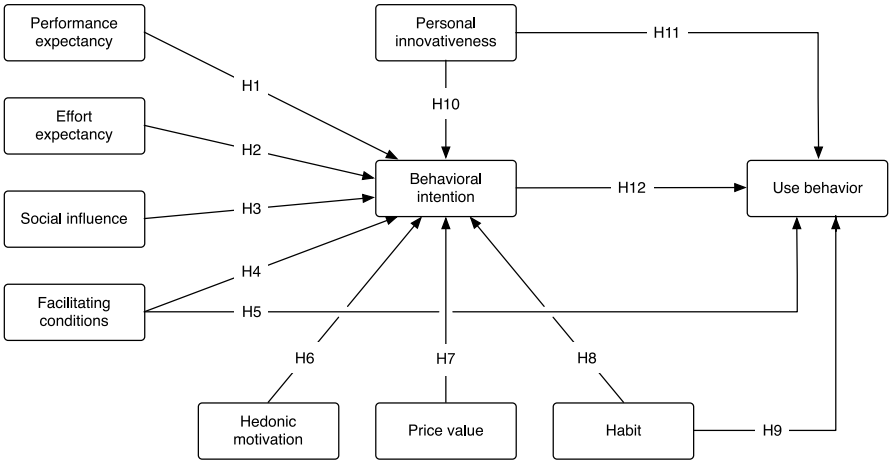

Fig. 1 A conceptual model investigating factors that may influence the adoption of ChatGPT among higher education students in the Nordic countries and the USA

Zoom (zoom.us), Webex (webex.com), and Google Meet (google.com/meet) for e-learning. The results showed that performance expectancy, social influence, hedonic motivation, learning value, and habit all demonstrated strong relationships with students' intention to use these platforms (Zacharis & Nikolopoulou, 2022). Another study in Spain examined students' behavioral intention to use the collaboration application Microsoft Teams (microsoft.com/teams) for e-learning. It was found that social influence had the greatest influence on students' intention to use the application, followed by the role of professors in shaping the perception of improvement (Garrido-Gutiérrez et al., 2023). Other research has also explored using extended UTAUT2 models to understand students' adoption of innovative educational technologies like meta-education platforms (Al-Adwan et al., 2024).

Adding to the cross-country perspective, a comparative study examined factors that influence university students' adoption of e-learning systems in Qatar and the USA (El-Masri & Tarhini, 2017). The results revealed that performance expectancy, hedonic motivation, habit, and trust were predictors of students' intention to adopt e-learning systems in both countries. The study found stronger positive relationships for effort expectancy and social influence in Qatar compared to the USA, where these relationships were less evident. Conversely, facilitating conditions showed a stronger positive relationship in the USA than in Qatar (El-Masri & Tarhini, 2017).

Demographic moderators such as age, gender, and experience from the UTAUT2 framework were excluded from the conceptual model in this study. This decision was based on empirical evidence and methodological considerations. First, recent studies on ChatGPT adoption in higher education have found that demographic moderators such as gender and education level (Strzelecki, 2023) and age and gender (Polyportis & Pahos, 2024) did not influence the relationships between predictors and dependent variables, despite their samples being demographically diverse. Second, a growing body of ChatGPT adoption research has examined student acceptance without incorporating

demographic moderators into theoretical models (Arif et al., 2024; Biloš & Budimir, 2024; Cai et al., 2023; Lai et al., 2023; Strzelecki, 2024). These studies have demonstrated that meaningful insights into ChatGPT adoption can be obtained by focusing on the direct relationships between key constructs. Third, the exclusion of moderators aligns with broader methodological trends in technology acceptance research, where studies increasingly favor more parsimonious models that omit moderators to facilitate the direct examination of relationships between constructs (Dwivedi et al., 2019, 2020). This approach has been prevalent in educational and AI contexts (An et al., 2023a, 2023b; Mohamed & Hassan, 2023; Tian & Yang, 2023). Additionally, the exclusion of moderators facilitates the broader application of the conceptual model across different settings and contexts (Habibi et al., 2023), which is particularly important given the study's focus on regional differences between the Nordic countries and the USA.

The UTAUT2 framework, extended with personal innovativeness, provides a theoretical foundation for examining ChatGPT adoption given its launch in late 2022. The framework guides the current investigation of factors influencing students' use of ChatGPT in higher education across the Nordic countries and the USA. The following sections discuss each construct in the UTAUT2 framework and develop the study's hypotheses.

2.2 Performance expectancy

In the UTAUT2 framework, performance expectancy is defined as the "degree to which using a technology will provide benefits to consumers in performing certain activities" (Venkatesh et al., 2012, p. 159). When applied to students' use of AI in higher education, performance expectancy can be understood as the extent to which they believe using ChatGPT will help them achieve better academic outcomes or enhance their productivity. Research has demonstrated that performance expectancy is an important predictor of behavioral intention to use AI (An et al., 2023; Cortez et al., 2024; Dahri et al., 2024).

An et al. (2023) showed that performance expectancy predicted behavioral intention to use AI in second language learning among junior and senior students. Another study examined students' academic use of communicational AI (CAI) as an educational tool (Cortez et al., 2024). Examples of CAIs include Grammarly (grammarly.com), ChatGPT (chat.openai.com), Quillbot (quillbot.com), and Jenni AI (jenni.ai). The findings revealed that performance expectancy had a direct effect on behavioral intention and an indirect effect on actual use (Cortez et al., 2024). Similarly, a cross-country study examined the determinants of higher education students' behavioral intention and use of AI tools (e.g., chatbots and machine learning applications) for academic support in Pakistan and Malaysia (Dahri et al., 2024). The data showed that performance expectancy positively influences behavioral intention in both samples, although it is negligibly weaker in the Pakistan sample. Based on existing research, this study puts forth the following hypothesis:

H1: Performance expectancy has a positive influence on the behavioral intention of students to adopt ChatGPT in higher education.

2.3 Effort expectancy

Effort expectancy is defined as the "degree of ease associated with consumers' use of technology" (Venkatesh et al., 2012, p. 159). In the context of AI in higher education, effort expectancy can be characterized as students' perceptions of the effort required to learn, interact with, and utilize ChatGPT in learning activities. Studies have demonstrated the extensive impact of effort expectancy on behavioral intention to use educational technologies (Altalhi, 2021; Chao, 2019; Sitar-Tăut, 2021), particularly AI (Kwak et al., 2022; Tran et al., 2021; Wu et al., 2022).

Studies have explored the factors influencing behavioral intention to use mobile learning (m-learning). For instance, Chao (2019) conducted a study among college students in Taiwan before the COVID-19 pandemic, while Sitar-Tăut (2021) carried out similar research during the global health crisis in a large European Union country. Despite the differences in geographical location, timing, and circumstances, both studies found that effort expectancy positively correlated with behavioral intention. Added to that, a study conducted in Saudi Arabia examined students' adoption of massive open online courses (MOOCs) for academic purposes (Altalhi, 2021). The results demonstrate that effort expectancy is crucial in determining students' behavioral intention to use MOOCs.

Within the realm of AI, research in China has found that effort expectancy is positively related to college students' willingness to accept AI-assisted learning environments (Wu et al., 2022). In contrast, a study in Vietnam examining medical students' intentions to adopt an AI-based diagnosis support system found that effort expectancy was positively associated with initial trust, but did not directly impact behavioral intention (Tran et al., 2021). Finally, Kwak et al. (2022) explored South Korean nursing students' intent to use AI-based healthcare technology. The study showed that effort expectancy influenced positive attitudes, which subsequently predicted behavioral intention to use medical tools driven by AI. Overall, research has demonstrated that minimizing the perceived difficulty of using a technological system can lead to higher utilization levels among users. Consistent with the mentioned results, the following hypothesis is proposed:

H2: Effort expectancy has a positive influence on the behavioral intention of students to adopt ChatGPT in higher education.

2.4 Social influence

In the educational context, research has found that students' choices to adopt or reject technological tools are often shaped by the opinions, behaviors, and recommendations of others in their social environment (Kabra et al., 2023; VanDerschaaf et al., 2023). Scholars refer to this phenomenon as social influence, which is broadly defined as "the extent to which consumers perceive that important others (e.g., family and friends) believe they should use a particular technology" (Venkatesh et al., 2012, p. 159). In this study, social influence represents the extent to which students perceive that classmates, faculty members, or other influential figures in their

social environment promote, stimulate, or assist them in interacting with and using ChatGPT.

Research has demonstrated that students' behavioral intention to adopt educational technologies is impacted by social influence. This impact has been observed across various technologies, including e-learning platforms (Zacharis & Nikolopoulou, 2022), MOOCs (Haron et al., 2021), and social media used as learning tools (Escobar-Rodríguez et al., 2014). Nevertheless, the impact of social influence on predicting behavioral intention to use educational technologies varies depending on the specific context and the technology being investigated. Some studies have found that social influence may not consistently have a positive effect on behavioral intention, as seen in instances of m-learning (Venkataraman & Ramasamy, 2018) and interactive whiteboards (Wong et al., 2013). Despite these mixed results, when it comes to AI in education, studies have shown that social influence plays a vital role in shaping students' behavioral intention to use AI-based technologies for learning purposes (Strzelecki, 2023; Tran et al., 2021; Wu et al., 2022). Considering the mentioned findings, this study proposes the following hypothesis:

H3: Social influence has a positive influence on the behavioral intention of students to adopt ChatGPT in higher education.

2.5 Facilitating conditions

Facilitating conditions can be defined as "consumers' perceptions of the resources and support available to perform a behavior" (Venkatesh et al., 2012, p. 159). While the construct of facilitating conditions was initially only connected to use behavior (Venkatesh et al., 2003), a meta-analysis by Dwivedi et al. (2019) suggested it should also be linked to behavioral intention. In higher education, facilitating conditions can be viewed as the availability of adequate resources, such as necessary knowledge, practical training, and support, which can directly influence students' intentions to use educational systems (Ahmed et al., 2022; Ain et al., 2016; Batucan et al., 2022). In this study, facilitating conditions signify students' perceived ability to access ChatGPT, and the availability of technical support and training for the AI-enabled chatbot. Although facilitating conditions sometimes fail to predict students' behavioral intention to use AI (An, Chai, Li, Zhou, & Yang, 2023; Weiqi et al., 2024), several studies have indicated that facilitating conditions can influence students' willingness to adopt AI technologies in educational settings (Foroughi et al., 2023; Habibi et al., 2023; Kwak et al., 2022; Paraskevi et al., 2023).

Foroughi et al. (2023) examined the adoption of ChatGPT among university students in Malaysia. They discovered that facilitating conditions positively impacted their behavioral intention to use the AI-based chatbot. Similarly, Habibi et al. (2023) found that facilitating conditions influenced Indonesian students' behavioral intention and use behavior regarding the adoption and utilization of ChatGPT in higher education. Moreover, a study conducted in South Korea explored nursing students' intention to adopt AI-based healthcare tools and found that facilitating conditions directly influenced their behavioral intention to use intelligent medical technologies

driven by AI (Kwak et al., 2022). Finally, a study in Greece investigated the factors influencing students' behavioral intention to adopt AI-powered chatbots in mobile contexts. The findings revealed that facilitating conditions positively impacted students' behavioral intention to use AI conversational agents on their smartphones (Paraskevi et al., 2023). Given the cited research, the following hypotheses were formulated:

H4: Facilitating conditions has a positive influence on the behavioral intention of students to adopt ChatGPT in higher education.
H5: Facilitating conditions has a positive influence on ChatGPT use behavior among students in higher education.

2.6 Hedonic motivation

Hedonic motivation is characterized as "the fun or pleasure derived from using a technology" (Venkatesh et al., 2012, p. 161). In higher education, hedonic motivation can be described as the extent to which students experience technologies to be entertaining, fun, and satisfying in a learning environment (Gulati et al., 2024; Hu et al., 2020). Specifically, in this study, hedonic motivation denotes the degree to which students perceive ChatGPT as enjoyable or fun in learning activities. Hedonic motivation has been shown to directly influence students' behavioral intention to adopt and engage with learning technologies in educational contexts, such as e-learning (Twum et al., 2022), m-learning (Al-Azawei & Alowayr, 2020; Moorthy et al., 2019), and learning coding (Cheon et al., 2022). Research has also established that hedonic motivation is vital in determining students' behavioral intention to adopt AI in higher education (Cai et al., 2023; Li & Qin, 2023; Vitezić & Perić, 2021).

Twum et al. (2022) demonstrated that hedonic motivation impacts university students' behavioral intention to adopt an online learning management system during the COVID-19 pandemic in Ghana. Al-Azawei and Alowayr (2020) and Moorthy et al. (2019) examined the impact of hedonic motivation on students' behavioral intention to use m-learning in Saudi Arabia, Iraq, and Malaysia, respectively. Both studies revealed that hedonic motivation positively influenced behavioral intention. Al-Azawei and Alowayr (2020) noted a more pronounced effect in Saudi Arabia than in Iraq, and Moorthy et al. (2019) identified hedonic motivation as one of the strongest predictors of behavioral intention among higher education students in Malaysia.

Concerning AI in higher education, two studies in China examined factors influencing students' intention to use AI-assisted learning tools. Cai et al. (2023) found that hedonic motivation influenced performance expectancy and perceived satisfaction, which positively impacted behavioral intention toward ChatGPT-assisted language learning. Similarly, Li and Qin (2023) discovered that hedonic motivation positively impacted medical students' intention to use medical AI systems, such as those for disease screening and diagnosis. Finally, Vitezić and Perić (2021) examined the willingness of students in Croatia to accept AI devices in the hospitality

industry. The researchers found that hedonic motivation impacted performance expectancy, which subsequently influenced emotions and, in turn, affected students' willingness to adopt these devices. Guided by the evidence from the reviewed literature, this study suggests the following hypothesis:

H6: Hedonic motivation has a positive influence on the behavioral intention of students to adopt ChatGPT in higher education.

2.7 Price value

Venkatesh et al. (2012) define price value as "consumers' cognitive tradeoff between the perceived benefits of the applications and the monetary cost for using them" (p. 161). The construct of price value is derived from the broader notion of perceived value, which is often considered a crucial indicator of user behavior (Al-Debei et al., 2013; Huang & Kao, 2015). In higher education, price value represents students' assessment of the financial costs associated with acquiring access to tools and online resources for learning purposes (Al-Azawei & Alowayr, 2020; Strzelecki, 2023). In this context, if students perceive that the benefits of using ChatGPT to support their learning activities and improve their academic outcomes are greater than the associated costs, then price value will positively affect their behavioral intention to use AI services.

Research has demonstrated that price value has a positive effect on the adoption of educational technologies, such as blended learning (Azizi et al., 2020), virtual reality (Shen et al., 2022), 3D interactive environments (Al-Adwan & Al-Debei, 2024), and AI (Wang & Zhang, 2023; Zhu et al., 2024). Other studies have found that price value had negligible or no influence on behavioral intention to use augmented reality (El-Masri & Tarhini, 2017; Faqih & Jaradat, 2021; Twum et al., 2022). Regarding AI, one study in China explored factors that influenced Generation Z's adoption of AI-assisted design tools. Wang and Zhang (2023) found that price value positively influenced the behavioral intention of students to use generative AI, while personal traits such as optimism, innovativeness, and curiosity also play important roles. Similarly, Zhu et al. (2024) discovered that price value positively affected university students' behavioral intention to use generative AI products. Based on this analysis, this study postulates the following hypothesis:

H7: Price value has a positive influence on the behavioral intention of students to adopt ChatGPT in higher education.

2.8 Habit

Habit is defined as "the extent to which people tend to perform behaviors automatically because of learning" (Limayem et al., 2007; Venkatesh et al., 2012, p. 161). According to habit theory, behaviors performed repeatedly over time can

become deeply entrenched and automatic through reinforcement (Wood & Neal, 2007). Building on previous work, habit is defined in this study as the degree to which students consistently and frequently engage with ChatGPT's functionalities over an extended period to foster favorable attitudes and increase the likelihood of incorporating ChatGPT into their learning process.

Studies have demonstrated a strong association between habit and digital learning technologies, such as m-learning (Arain et al., 2019; Sitar-Taut & Mican, 2021), learning management systems such as Moodle (Zwain, 2019), and Google Classroom (Bervell et al., 2021), chatbots (e.g., Siri, Alexa, Google Assistant) (Almahri et al., 2024), open educational resources (e.g., Open Education Consortium, MIT OpenCourseWare, MOOCs) (Jung & Lee, 2020), and AI (Almahri et al., 2024; Lavidas et al., 2024; Salifu et al., 2024; Sebastián et al., 2022). However, not all studies have found an influence of habit on behavioral intention. One study on blended learning suggested that other factors may have a prominent role in shaping students' adoption of this hybrid learning approach (Rudhumbu, 2022).

Regarding AI in educational settings, an investigation into the adoption of ChatGPT among 306 economics students in Ghanaian universities found that habit affected students' behavioral intentions and actual usage (Salifu et al., 2024). Similarly, a study examining the adoption of conversational AI, or intelligent virtual assistants, among 304 students in Spain discovered that habit played a crucial role in their adoption (Sebastián et al., 2022). Moreover, an exploration of the factors influencing 197 Greek students' intention to use and actual use of AI services (e.g., ChatGPT, Copilot, Stable Diffusion) in higher education identified habit as a determinant (Lavidas et al., 2024). These studies underscore the importance of habit in shaping student engagement with AI technologies across educational contexts and countries. Drawing from these insights, this study presents the following hypotheses:

H8. Habit has a positive influence on the behavioral intention of students to adopt ChatGPT in higher education.
H9. Habit has a positive influence on ChatGPT use behavior among students in higher education.

2.9 Personal innovativeness

Individuals respond to innovation in different ways. Some individuals are more willing to embrace novel ideas and practices, whereas others are more cautious about changing their practices (Fan et al., 2020; Yi et al., 2006). The desire to experiment with and adopt innovation is a personality trait known as personal innovativeness (Ciftci et al., 2021). In information technology, personal innovativeness is defined as "the willingness of an individual to try out any new information technology" (Agarwal & Prasad, 1998, p. 206). Expanding on this definition, personal innovativeness has been characterized as "the willingness to adopt [the] latest technological gadgets, or risk-taking propensity, which might be attached to trying new features and advancements in the domain of IT" (Farooq et al., 2017, p. 334). In the context of AI

and this study, personal innovativeness is defined as students' willingness to adopt emerging AI services, such as ChatGPT, and their self-assurance in acquiring the necessary skills to leverage them.

Research has consistently shown that students' innovativeness is essential to their willingness to adopt new technologies. Studies have reported a positive influence of personal innovativeness on the behavioral intention to use educational technologies, such as MOOCs (Amid & Din, 2021; Shah & Khanna, 2024), learning through metaverse technology (Al-Adwan & Al-Debei, 2024), learning capture systems (i.e., digital recording of lectures for online streaming) (Farooq et al., 2017), e-textbooks (Ngafeeson & Sun, 2015), mobile learning (Chen, 2022; Sidik & Syafar, 2020), and learning management systems such as Moodle (Pinho et al., 2021).

The impact of personal innovativeness on AI chatbots, such as ChatGPT, particularly in educational contexts, has been a topic of interest. Some studies have found that personal innovativeness positively influences the adoption of AI chatbots in China (Weiqi et al., 2024), specifically ChatGPT in Bangladesh (Arif et al., 2024), Croatia (Biloš & Budimir, 2024), South Korea (Jo, 2023), and Poland (Strzelecki, 2023). However, other studies have reported conflicting results and have shown that personal innovativeness does not have an impact on the adoption of e-learning (Gunasinghe et al., 2019) or ChatGPT (Foroughi et al., 2023; Strzelecki, 2024). These mixed findings highlight the need for further research to understand the role of personal innovativeness in ChatGPT adoption. Informed by the current research, this study advances the following hypotheses:

H10: Personal innovativeness has a positive influence on the behavioral intention of students to adopt ChatGPT in higher education.
H11: Personal innovativeness has a positive influence on ChatGPT use behavior among students in higher education.

2.9.1 Behavioral intention

Behavioral intention has been shown to predict use behavior in the UTAUT2 framework (Dwivedi et al., 2011; Venkatesh et al., 2012). The construct is typically defined as "an individual's subjective probability that he or she will perform a specified behavior" (Fishbein & Ajzen, 1975, p. 288). Research has since expanded on the previous definition by describing behavioral intention as "a reflection of an individual's internal schema of beliefs" that "[…] does not represent the external factors that can influence the performance of a behavior" (Venkatesh et al., 2008, p. 485). When applied to information technology, behavioral intention takes on a more specific meaning and refers to an individual's intention to use a particular technology for different tasks (Ain et al., 2016). In this study, behavioral intention measures the level of commitment students convey towards engaging with AI technologies, specifically ChatGPT, to achieve their educational objectives and goals.

Studies have found a strong association between students' behavioral intention and actual technology usage within educational contexts, such as learning management systems (Abbad, 2021; Al-Mamary et al., 2023), mobile language learning (García Botero et al., 2018; Hoi, 2020), as well as animation and storytelling (Suki

& Suki, 2017). Similarly, researchers have unveiled this relationship across AI applications, ranging from learning AI in healthcare education (Li et al., 2022) to learning experiences facilitated by AI chatbots (Rahim et al., 2022), specifically ChatGPT (Acosta-Enriquez et al., 2024; Gulati et al., 2024; Jo, 2023; Salifu et al., 2024).

Medical students in China who exhibited strong behavioral intentions to learn AI for clinical practice had better learning outcomes (Li et al., 2022). In Malaysia, researchers discovered that behavioral intention is an essential factor that positively influences the actual use of AI-based chatbots in higher education institutions (Rahim et al., 2022). The impact of behavioral intention on actual use has also been observed in studies focusing on the adoption of ChatGPT across diverse countries, including Peru (Acosta-Enriquez et al., 2024), India (Gulati et al., 2024), South Korea (Jo, 2023), and Ghana (Salifu et al., 2024). The collective findings from the reviewed research guide this study to suggest the following hypothesis:

H12: Behavioral intention has a positive influence on ChatGPT use behavior among students in higher education.

2.9.2 Use behavior

Use behavior is the final endogenous construct in the UTAUT2 framework that seeks to predict how individuals use technology. The seminal work of Venkatesh et al. (2012) did not specify the measurement of use behavior. Following Strzelecki (2024), this study operationalized the use of ChatGPT by adopting a seven-point scale, ranging from "never" to "several times a day" to capture the varying levels of user engagement with the AI-service.

3 Methodology

3.1 Instrumentation

An online questionnaire with four sections was created to evaluate the proposed hypotheses. Unless otherwise stated, the questionnaire content was identical for the Nordic and USA samples. The first section presented an overview of the study, outlined ethical considerations, and requested participants' informed consent. Participants were asked to give informed consent by indicating "yes" or "no". Selecting the former would allow a participant to proceed with the survey, whereas choosing the latter would end their participation. The second section collected demographic information, such as age, gender, education level, and country of residence (for Nordic participants only). The third section of the questionnaire comprised 33 items designed to measure nine constructs related to university students' behavioral intention toward using ChatGPT. The constructs, adapted from previous research (Agarwal & Prasad, 1998; Strzelecki, 2023; Venkatesh et al., 2003, 2012), included performance expectancy, effort expectancy, social influence, facilitating conditions, hedonic motivation, price value, habit, personal innovativeness, behavioral intention, and use behavior. The constructs and their associated items are described in Table 1.

As depicted, performance expectancy, effort expectancy, facilitating conditions, habit, and personal innovativeness were operationalized with four items each. Social influence, hedonic motivation, price value, and behavioral intention each consisted of three items, whereas the use behavior had one. Each item contained a statement measured on a seven-point Likert-type scale ranging from 1 (strongly disagree) to 7 (strongly agree). The construct of use behavior was measured using a single item (i.e., "How often do you use ChatGPT?") with response options on a seven-point scale ranging from 1 (never) to 7 (several times a day). As Johnson et al. (2014) argued, measuring actual system use can be challenging because the necessary data are difficult to obtain and often unavailable to researchers. The fourth section consisted of a debriefing that thanked the participants for their valuable contributions.

Before distributing the questionnaire, two experts in higher education were invited to assess its face and content validity. Based on their feedback, the wording and sentence structure of select statements were improved in terms of clarity. In addition, a pilot study was conducted with 80 participants using purposive sampling on Prolific. It is important to note that these participants were excluded from the main study. Participants in the pilot study consisted of 40 females, 38 males, and 2 non-binary/third gender, with an age range of 18 to 54 years ($M = 25.55$; $SD = 6.08$). The findings of the pilot study demonstrated overall satisfactory reliability (Cronbach's alpha > 0.7; PE = 0.850, EE = 0.774, SI = 0.893, FC = 0.639, HM = 0.915, PV = 0.940, HB = 0.767, PI = 0.892, BI = 0.872) and convergent validity (AVE > 0.5; PE = 0.690, EE = 0.600, SI = 0.823, FC = 0.525, HM = 0.852, PV = 0.881, HB = 0.586, PI = 0.749, BI = 0.697) of the measurement scale, and no additional changes were required.

3.2 Data collection

This study employed an intercontinental research design to collect data from students in five Nordic countries and the USA. The survey was created using the Qualtrics Core XM service and disseminated online to participants through the Prolific recruitment platform. Research has shown that Prolific participants outperform those from MTurk and Qualtrics in passing attention checks, providing meaningful responses, and following instructions (Douglas et al., 2023; Peer et al., 2022). Purposive sampling was used to select participants who met specific pre-screening criteria aligned with the research objectives of this study (Etikan et al., 2016; Tongco, 2007).

Pre-screening criteria were applied before distributing the questionnaire on Prolific (McKee et al., 2023). For the Nordic sample, participants had to reside in one of the five Nordic countries, be enrolled as students, and have completed at least two previous studies with a 95% approval rate or above. The lower threshold of two previous studies was set because of the relatively low number of available participants. For the USA sample, participants had to reside in the USA, be current students, and have completed at least 20 previous studies with a 95% approval rate or higher. Upon completing the data collection, a total of 649 participants responded to the questionnaire (334 from the Nordic countries and 315 from the USA).

Table 1 Measurement scale

Construct	Item	Statement	Source
Performance expectancy	PE1	I believe that ChatGPT is useful in my studies	Venkatesh et al., (2003, 2012)
	PE2	Using ChatGPT increases your chances of achieving important things in your studies	
	PE3	Using ChatGPT helps you get tasks and projects done faster in your studies	
	PE4	Using ChatGPT increases your productivity in your studies	
Effort expectancy	EE1	Learning how to use ChatGPT is easy for me	Venkatesh et al., (2003, 2012)
	EE2	My interaction with ChatGPT is clear and understandable	
	EE3	I find ChatGPT easy to use	
	EE4	It is easy for me to become skillful at using ChatGPT	
Social influence	SI1	People who are important to me think I should ChatGPT	Venkatesh et al. (2012)
	SI2	People who influence my behavior believe that I should use ChatGPT	
	SI3	People whose opinions I value prefer me to use ChatGPT	
Facilitating conditions	FC1	I have the resources necessary to use ChatGPT	Venkatesh et al., (2003, 2012)
	FC2	I have the knowledge necessary to use ChatGPT	
	FC3	ChatGPT is compatible with technologies I use	
Hedonic motivation	HM1	Using ChatGPT is fun	Venkatesh et al. (2012)
	HM2	Using ChatGPT is enjoyable	
	HM3	Using ChatGPT is very entertaining	
Price value	PV1	ChatGPT is reasonably priced	Venkatesh et al. (2012)
	PV2	ChatGPT is good value for the money	
	PV3	At the current price, ChatGPT provides a good value	
Habit	HB1	The use of ChatGPT has become a habit for me	Venkatesh et al. (2012)
	HB2	I am addicted to using ChatGPT	
	HB3	I must use ChatGPT	
	HB4	Using ChatGPT has become natural for me	
Personal innovativeness	PI1	I like experimenting with new information technologies	Agarwal and Prasad (1998)
	PI2	If I heard about a new information technology, I would look for ways to experiment with it	

Table 1 (continued)

Construct	Item	Statement	Source
	PI3	Among my family/friends, I am usually the first to try out new information technologies	
	PI4	In general, I do not hesitate to try out new information technologies	
Behavioral intention	BI1	I intend to continue using ChatGPT in the future	Venkatesh et al. (2012)
	BI2	I will always try to use ChatGPT in my studies	
	BI3	I plan to continue to use ChatGPT frequently	
Use behavior	UB	How often do you use ChatGPT? Never; Once a month; Several times a month; Once a week; Several times a week; Once a day; Several times a day	Strzelecki (2023)

All items, except for use behavior, were measured on a 7-point Likert-type scale, where 1 = strongly disagree, 2 = disagree, 3 = somewhat disagree, 4 = neither agree nor disagree, 5 = somewhat agree, 6 = agree, and 7 = strongly agree

3.3 Participants

G*Power analysis was used to determine the minimum sample size (Faul et al., 2007). This calculation was based on statistical power considerations that factored in the desired effect size, probability level, power level, and number of predictors. With an effect size of 0.15, 0.05 probability level, 95% desired statistical power, and eight predictors, the analysis revealed that a minimum of 160 participants was required (Champely et al., 2017). Initial data analysis using SPSS identified and excluded missing values and outliers. The final sample consisted of 586 participants, 296 from the Nordic countries and 290 from the USA. Both samples individually exceeded the minimum required sample size. The demographic characteristics of the participants in both regions are presented in Table 2.

The age range for the participants from the Nordic countries was 18–59 years ($M = 26.28$; $SD = 6.45$), while for the USA it was 18–60 years ($M = 29.00$; $SD = 8.14$). Most participants in the Nordic sample were from Sweden ($n = 150$), followed by Norway ($n = 58$), Finland ($n = 50$), Denmark ($n = 34$), and Iceland ($n = 4$).

3.4 Data preparation and analysis

The data were evaluated for normality through the assessment of skewness and kurtosis. Data are generally considered normally distributed when skewness and kurtosis values fall within the range of −2 to + 2 (Byrne, 2010; Hair et al., 2022). As presented in Table 3, all skewness and kurtosis values fell within the recommended range, indicating that the data met the assumption of normality.

Table 2 Demographic characteristics of participants

	Nordic ($N = 296$)		USA ($N = 290$)	
Variables	Frequency	Percent	Frequency	Percent
Gender				
Female	137	46.3%	144	49.7%
Male	155	52.4%	141	48.6%
Non-binary / third gender	4	1.3%	5	1.7%
Age groups				
18–27 years*	200	67.6%	157	54.1%
28–43 years**	88	29.7%	112	38.6%
44 + years***	8	2.7%	21	7.3%
Education				
Two-year university degree	62	20.9%	89	30.7%
Bachelor's degree	140	47.3%	159	54.8%
Master's degree	87	29.4%	35	12.4%
Doctorate degree	7	2.4%	6	2.1%

The age groups are categorized as Millennials*, Generation Z**, and Generation X*** according to the generational definitions by Dimock (2019)

For the analyses, this study adopted partial least squares structural equation modeling (PLS-SEM) using SmartPLS 4 (version 4.1.0.6) to investigate the proposed hypotheses (Ringle et al., 2024). PLS-SEM was selected because of its high efficiency in parameter estimation and its robustness in handling complex models (Hair et al., 2022). PLS-SEM procedures, including measurement and structural model assessments, were conducted to examine the conceptual model in this study. The quadratic effect was assessed to test for potential nonlinearities in the relationships between constructs. Furthermore, the finite-mixture partial least squares (FIMIX-PLS) procedure was used to assess the unobserved heterogeneity of the data (Sarstedt et al., 2020). Finally, an importance-performance map analysis (IPMA) was employed to extend the results of PLS-SEM by considering the performance of each construct (Ringle & Sarstedt, 2016).

3.5 Ethical considerations

This study adhered to the four ethical principles set forth by the Swedish Research Council (Stafström, 2017): secrecy, professional secrecy, anonymity, and confidentiality. Before taking the questionnaire, participants were given an overview of the study's objectives and were asked to consent by choosing to voluntarily participate in the study or decline participation. They were assured that all data management would follow the mentioned ethical standards. To ensure the security of the collected data, they were stored on the Qualtrics Core XM service and were only accessible to the researchers directly involved in the study. Participants were also informed that their data would be confidential and not disclosed to any third party.

4 Results and analysis

4.1 Usage patterns across age groups

It is important to understand the usage patterns of ChatGPT across different age groups and regions, as this provides a context for interpreting the subsequent analyses in the measurement and structural models. The findings indicate distinct usage patterns across age groups and regions (see Table 4). In the Nordic sample, students aged 18–27 years ($n = 200$), who represent the largest group in the sample, the most common usage pattern was "Several times a week" (36.0%), followed by "Several times a month" (16.0%). For students aged 28–43 years ($n = 88$), the usage was more evenly distributed between "Several times a day" and "Several times a week" (26.1%). In the USA sample, students in the 18–27 age group ($n = 157$) showed different patterns. While "Several times a week" remained the most common frequency (26.1%), the overall distribution was more balanced across frequency categories. Students aged 28–43 years ($n = 112$) demonstrated a similar pattern, with "Several times a week" being the most common frequency (28.6%).

Table 3 Mean, standard deviation, skewness, and kurtosis for the constructs in the Nordic and USA models

		Nordic				USA			
Construct	Item	M	SD	Skew	Kur	M	SD	Skew	Kur
Performance expectancy	PE1	5.46	1.240	−1.102	1.384	5.19	1.454	−0.885	0.303
	PE2	4.76	1.459	−0.638	−0.153	4.74	1.572	−0.362	−0.613
	PE3	5.44	1.189	−0.996	1.563	5.29	1.379	−0.827	0.200
	PE4	5.18	1.288	−0.740	0.247	5.01	1.492	−0.698	0.051
Effort expectancy	EE1	5.86	0.934	−0.934	1.243	5.79	1.022	−1.007	1.929
	EE2	5.52	1.054	1.054	1.300	5.66	1.047	−0.927	1.423
	EE3	5.89	0.940	0.940	1.671	5.97	0.928	−0.959	1.167
	EE4	5.32	1.074	1.074	0.379	5.58	1.083	−0.849	1.181
Social influence	SI1	3.63	1.472	−0.161	−0.487	3.92	1.494	−0.014	−0.330
	SI2	3.67	1.428	−0.161	−0.311	3.88	1.536	−0.038	−0.489
	SI3	3.58	1.424	−0.034	−0.345	3.78	1.512	0.035	−0.335
Facilitating conditions	FC1	5.95	0.818	−0.998	1.864	6.00	0.774	−0.502	0.221
	FC2	5.75	1.004	−0.877	0.898	5.82	0.980	−1.094	1.908
	FC3	5.43	1.165	−1.083	1.811	5.71	0.998	−0.993	1.352
	FC4	4.76	1.443	−0.583	−0.234	5.04	1.276	−0.806	0.454
Hedonic motivation	HM1	5.28	1.155	−0.723	0.597	5.24	1.252	−0.476	−0.092
	HM2	5.16	1.127	−0.710	0.639	5.37	1.164	−0.700	0.567
	HM3	4.93	1.271	−0.471	−0.322	5.13	1.294	−0.503	−0.211
Price value	PV1	4.69	1.470	−0.237	−0.473	5.03	1.315	−0.321	−0.311
	PV2	4.87	1.453	−0.448	−0.119	5.18	1.251	−0.390	−0.029
	PV3	4.97	1.379	−0.359	−0.288	5.21	1.243	−0.434	−0.208
Habit	HB1	3.71	1.836	0.062	−1.184	3.60	1.898	0.181	−1.161
	HB2	2.10	1.363	1.274	1.074	2.22	1.491	1.427	1.474
	HB3	2.46	1.605	0.993	0.089	2.70	1.706	0.843	−0.325
	HB4	4.18	1.648	−0.289	−0.830	4.24	1.788	−0.446	−0.838
Personal innovativeness	PI1	5.58	1.167	−0.804	0.535	5.55	1.170	−1.052	1.419
	PI2	5.16	1.313	−0.692	−0.036	5.25	1.256	−0.679	0.514
	PI3	5.26	1.512	−0.769	−0.258	4.93	1.574	−0.572	−0.521
	PI4	5.37	1.413	−0.854	0.163	5.29	1.382	−0.815	0.146
Behavioral intention	BI1	5.61	1.129	−1.182	1.650	5.47	1.417	−1.282	1.650
	BI2	3.92	1.689	−0.030	−0.955	3.85	1.860	0.103	−1.078
	BI3	5.04	1.478	−0.869	0.259	4.89	1.643	−0.675	−0.300
Use behavior	UB	3.50	1.739	0.206	−0.886	3.84	1.813	0.138	−0.985

M, Mean; *SD*, Standard deviation

While the 44 + age group showed distinct patterns, the findings should be interpreted with caution due to the small sample sizes ($n = 8$ for Nordic and $n = 21$ for USA). In the Nordic sample, this group displayed a polarized usage pattern of 50% using ChatGPT "Several times a week," and 37.5% never used it.

Table 4 ChatGPT usage frequency by age groups by region (%)

Region	Age groups	STD	OD	STW	OW	STM	OM	Never
Nordic	18–27 years	15.5	6.0	36.0	8.5	16.0	14.5	3.5
	28–43 years	26.1	10.2	26.1	11.4	15.9	8.0	2.3
	44 + years	0.0	0.0	50.0	12.5	0.0	0.0	37.5
USA	18–27 years	8.9	10.8	26.1	14.6	15.3	14.6	9.6
	28–43 years	16.1	8.9	28.6	8.0	14.3	15.2	8.9
	44 + years	19.0	4.8	42.9	19.0	4.8	4.8	4.8

Values represent percentages within each age group. *STD*, Several times a day; *OD*, Once a day; *STW*, Several times a week; *OW*, Once a week; *STM*, Several times a month; *OM*, Once a month

In contrast, USA students in this age group showed more frequent usage, with 42.9% using it "Several times a week" and only 4.8% never used it. These results suggest that frequent usage is common across all age groups, but there are some regional differences in adoption patterns.

4.2 Common method bias

This study addressed the issue of common method bias (CMB), a systematic error that can occur when shared method variance artificially inflates relationships among observable variables (Podsakoff et al., 2024). Following Podsakoff et al. (2003), this study employed Harman's single-factor test to evaluate CMB. In the Nordic dataset, the first factor explained 29.29% of the variance, whereas in the USA dataset, it accounted for 39.56%. Both figures are below the 50% threshold (Podsakoff et al., 2003), indicating a low probability of CMB. Additionally, this study evaluated the variance inflation factor (VIF) scores to mitigate potential CMB. VIF can serve as a supplementary measure for detecting multicollinearity among predictor constructs and offers valuable insights into potential CMB when VIF scores exceed normal thresholds. For the Nordic and the USA datasets, all VIF scores were below the broadly accepted threshold of 5 (Hair et al., 2022) (see Table 5). These findings demonstrate that CMB is unlikely to affect the observed relationships between the included constructs.

4.3 Measurement model

The PLS-SEM algorithm was used to estimate the model by employing default initial weights and a maximum of 3000 iterations (Ringle et al., 2024). The reliability and validity of the measurement model were assessed using Cronbach's alpha (α), composite reliability (ρ_A and ρ_C), and average variance extracted (AVE) for each construct. Additionally, individual item loadings were analyzed to evaluate the internal consistency of the constructs. Item loadings greater than 0.708 indicate that the constructs explain more than 50% of the variance in the indicator, which is considered an acceptable level of item reliability (Hair et al., 2022). As presented in Table 5, all item loadings, except for FC2 in the Nordic sample and FC4 in both samples, were above the recommended threshold. Upon closer examination, the loading of FC2 was below the threshold, but not critically low. Based on the recommendations by Hair et al. (2022) for item reliability, only FC4 was removed from further processing in the model because of its suboptimal loading in both samples.

Cronbach's alpha is a measure of internal consistency reliability with acceptable values ranging from 0.7 or above. Likewise, composite reliability serves as an additional indicator of a model's reliability and has thresholds similar to Cronbach's alpha, ranging from 0.70 to 0.95 (Hair et al., 2022). Convergent validity, which assesses the degree to which a measure positively correlates with alternative

Table 5 Item loading and variance inflation factor for the constructs in the Nordic and USA models

Construct	Item	Nordic Loading	VIF	USA Loading	VIF
Performance expectancy	PE1	0.889	2.438	0.923	3.624
	PE2	0.824	1.951	0.898	3.080
	PE3	0.844	2.288	0.845	2.432
	PE4	0.873	2.503	0.921	3.578
Effort expectancy	EE1	0.859	2.259	0.878	2.930
	EE2	0.786	1.611	0.851	2.178
	EE3	0.865	2.182	0.856	2.428
	EE4	0.746	1.543	0.851	2.295
Social influence	SI1	0.933	3.541	0.942	3.665
	SI2	0.897	3.017	0.926	3.366
	SI3	0.925	2.725	0.915	2.941
Facilitating conditions	FC1	0.777	1.532	0.782	1.683
	FC2	0.678	1.397	0.823	1.556
	FC3	0.817	1.248	0.835	1.490
	FC4	0.396	1.062	0.382	1.054
Hedonic motivation	HM1	0.922	3.036	0.942	4.123
	HM2	0.913	2.365	0.941	3.676
	HM3	0.871	2.524	0.903	2.871
Price value	PV1	0.927	3.841	0.905	2.924
	PV2	0.960	4.912	0.933	3.616
	PV3	0.945	4.240	0.934	3.522
Habit	HB1	0.880	2.358	0.897	3.048
	HB2	0.755	1.802	0.792	1.959
	HB3	0.710	1.639	0.804	2.089
	HB4	0.844	2.095	0.876	2.671
Personal innovativeness	PI1	0.890	2.616	0.887	2.851
	PI2	0.892	2.580	0.894	2.823
	PI3	0.797	1.956	0.820	1.907
	PI4	0.850	2.104	0.872	2.402
Behavioral intention	BI1	0.863	2.248	0.876	2.455
	BI2	0.848	1.866	0.875	2.181
	BI3	0.925	2.860	0.937	3.417
Use behavior	UB	1.000	1.000	1.000	1.000

VIF, Variance inflation factor

measures of the same construct, was evaluated using AVE, with values of 0.5 or higher generally considered acceptable (Becker et al., 2023; Russo & Stol, 2021; Shmueli et al., 2019). All constructs met the quality criteria presented in Table 6, except for facilitating conditions in the Nordic sample, which had a value for Cronbach's alpha that was negligibly below the recommended threshold. However, as this value was not critically low, it was deemed acceptable for further analysis.

Based on the AVE results, facilitating conditions showed the lowest values, 0.600 for the Nordic model and 0.671 for the USA model, explaining 60.0% and 67.1% of the variance, respectively. The highest AVE was observed for price value in the Nordic model at 0.891, accounting for 89.1% of the variance, while hedonic motivation led in the USA model at 0.863, explaining 86.3% of the variance.

Furthermore, discriminant validity was assessed using the heterotrait-monotrait (HTMT) ratio of correlations method (Henseler et al., 2015). This assessment ensures that each construct is empirically unique and captures aspects that are not represented by other constructs in the model (Hair et al., 2022). A threshold of 0.90 is recommended for the HTMT ratio when assessing conceptually similar constructs, while a more conservative 0.85 is suggested for distinct constructs (Henseler et al., 2015). Values exceeding these levels could indicate discriminant validity issues. The results in Table 7 demonstrate that all HTMT values fall below the conservative threshold of 0.85, providing evidence for discriminant validity among the constructs in the proposed model.

To further assess discriminant validity, the $HTMT_{inference}$ statistic was calculated through bootstrapping with 5000 samples with a one-tailed test type. This criterion evaluates whether the HTMT values fall below the critical level of 0.9, with values under this threshold indicating evidence of discriminant validity (Franke & Sarstedt, 2019). The analysis revealed that all values of the upper bound (95%) for the confidence intervals bias corrected were below 0.9. The highest observed values were 0.871 for the Nordic model and 0.891 for the USA model, both occurring between performance expectancy and behavioral intention. All other construct pairs demonstrated values of 0.864 or lower for the Nordic model and 0.862 or lower for the USA model.

Table 6 Construct reliability and validity for the constructs in the Nordic and USA models

	Cronbach's alpha (α)		Composite reliability (ρ_A)		Composite reliability (ρ_C)		Average variance extracted (AVE)	
Construct	Nordic	USA	Nordic	USA	Nordic	USA	Nordic	USA
PE	0.880	0.922	0.891	0.927	0.918	0.944	0.736	0.810
EE	0.831	0.890	0.837	0.893	0.888	0.924	0.665	0.751
SI	0.908	0.917	0.937	0.922	0.942	0.948	0.843	0.858
FC	0.681	0.760	0.723	0.778	0.818	0.859	0.600	0.671
HM	0.887	0.921	0.919	0.933	0.929	0.950	0.814	0.863
PV	0.939	0.918	0.961	0.922	0.961	0.948	0.891	0.859
HB	0.815	0.868	0.849	0.884	0.876	0.909	0.640	0.715
PI	0.882	0.891	0.901	0.895	0.918	0.924	0.737	0.754
BI	0.852	0.875	0.860	0.880	0.911	0.923	0.773	0.801

PE, Performance expectancy; *EE*, Effort expectancy; *SI*, Social influence; *FC*, Facilitating conditions; *HM*, Hedonic motivation; *PV*, Price value; *HB*, Habit; *PI*, Personal innovativeness; *BI*, Behavioral intention

Table 7 Discriminant validity and HTMT values for the constructs in the (a) Nordic and (b) USA models

	BI	PE	EE	SI	FC	HM	PV	HB	PI	UB
(a) Nordic										
BI	—									
PE	0.822	—								
EE	0.477	0.415	—							
SI	0.506	0.467	0.179	—						
FC	0.416	0.392	0.639	0.240	—					
HM	0.475	0.392	0.441	0.229	0.263	—				
PV	0.364	0.369	0.340	0.240	0.391	0.225	—			
HB	0.809	0.688	0.429	0.525	0.288	0.370	0.213	—		
PI	0.349	0.359	0.500	0.138	0.313	0.352	0.081	0.377	—	
UB	0.705	0.558	0.306	0.334	0.243	0.270	0.238	0.696	0.338	—
(b) USA										
BI	—									
PE	0.847	—								
EE	0.577	0.519	—							
SI	0.621	0.502	0.274	—						
FC	0.556	0.476	0.793	0.291	—					
HM	0.593	0.504	0.623	0.415	0.520	—				
PV	0.488	0.483	0.434	0.339	0.465	0.435	—			
HB	0.820	0.684	0.467	0.649	0.378	0.534	0.381	—		
PI	0.625	0.514	0.562	0.392	0.560	0.540	0.282	0.444	—	
UB	0.670	0.500	0.401	0.455	0.381	0.344	0.251	0.649	0.398	—

PE, Performance expectancy; *EE*, Effort expectancy; *SI*, Social influence; *FC*, Facilitating conditions; *HM*, Hedonic motivation; *PV*, Price value; *HB*, Habit; *PI*, Personal innovativeness; *BI*, Behavioral intention; *UB*, Use behavior

4.4 Structural model

4.4.1 Assessment of nonlinearity and unobserved heterogeneity

This study assessed the quadratic effect to test for potential nonlinearities (Sarstedt et al., 2020). The results of bootstrapping with 5000 samples revealed quadratic effects between habit and behavioral intention in the USA model ($f^2 = 0.052$), and between habit and behavioral intention ($f^2 = 0.037$) and use behavior ($f^2 = 0.012$) in the Nordic model. All other quadratic effects were insignificant. Given these results, further research is needed to examine the relationship between habit and use behavior in the USA, particularly to understand why a linear relationship persists. Concerning habit, as students begin to use ChatGPT more frequently, their behavioral intention and use behavior can increase significantly. This is because of the growing familiarity and perceived benefits that accompany repeated use. However, after reaching a certain level of habitual use, behavioral intention and use behavior can plateau and potentially decrease. Students who already use ChatGPT regularly

may not experience the same increase in behavioral intention or use behavior as the AI service becomes a routine part of their study habits.

Another robustness check was conducted by examining unobserved heterogeneity. This concept refers to differences across unobserved subgroups within a dataset, which can threaten the validity of the PLS-SEM results (Sarstedt et al., 2020). The finite-mixture partial least squares (FIMIX-PLS) procedure was used to assess the unobserved heterogeneity of the collected data. Before running the FIMIX-PLS, it was essential to establish the highest number of segments to extract and the least sample size necessary to estimate each segment. An effect size of 0.15 with a power level of 80% produced a required sample size of 109, which allowed the extraction of up to five segments. The FIMIX-PLS procedure was conducted for one to five segments for both the Nordic and USA models using the stop criterion ($10^{-10} = 1.0E-10$), the maximum number of iterations (5000), and the number of repetitions (10) (see Table 8).

For the Nordic model, AIC_3 indicated a four-segment solution, whereas CAIC pointed to a five-segment solution. Furthermore, AIC_4 and BIC pointed to different numbers of segments, AIC_4 to a four-segment solution and BIC to a one-segment solution. For the USA model, a similar pattern emerged with AIC_3, which indicated a five-segment solution, whereas CAIC pointed to a one-segment solution. Additionally, AIC_4 and BIC pointed to different numbers of segments, AIC_4 to a three-segment solution and BIC to a one-segment solution. However, MDL_5 pointed to a one-segment solution in both models. Overall, the analyses do not clearly indicate a

Table 8 Fit indices for one- to five-segment solutions for the Nordic and USA models

		Number of segments				
Model	Criteria	1	2	3	4	5
Nordic	AIC	1181.850	1172.776	1147.461	1089.426	**1079.339**
	AIC_3	1195.850	1201.776	1191.461	**1148.426**	1153.339
	AIC_4	1209.850	1230.776	1235.461	**1207.426**	1352.426
	BIC	**1233.515**	1279.796	1309.837	1307.158	1426.426
	CAIC	1247.515	1308.796	1353.837	1366.158	**1227.339**
	MDL_5	**1552.175**	1939.878	2311.340	2650.083	3036.772
	EN	NA	0.616	**0.920**	0.896	0.874
USA	AIC	1114.024	1093.379	1046.630	1035.755	**998.306**
	AIC_3	1128.024	1122.379	1090.630	1094.755	**1072.306**
	AIC_4	1142.024	1151.379	**1134.630**	1153.755	1146.306
	BIC	**1165.403**	1199.805	1208.105	1252.278	1269.877
	CAIC	**1179.403**	1228.805	1252.105	1311.278	1343.877
	MDL_5	**1482.916**	1857.511	2206.004	2590.370	2948.162
	EN	NA	0.743	**0.868**	0.793	0.807

AIC,: Akaike's information criterion; *AIC_3*, Modified AIC with factor 3; *AIC_4*, Modified AIC with factor 4; *BIC*, Bayesian information criteria; *CAIC*, Consistent AIC; *MDL_5*, Minimum description length with factor 5; *EN*, Entropy statistic; *NA*, Not available; Numbers highlighted in bold show the best outcome per segment

definitive segmentation because (1) AIC_3 and CAIC pointed to different segments, and (2) AIC_4 indicated a different segment compared to BIC and MDL_5. Therefore, unobserved heterogeneity was not identified as critical.

4.4.2 Evaluation of model fit

Confirmatory factor analysis (CFA) was conducted using IBM SPSS AMOS 29.0 to examine the construct validity of the conceptual model. Several measures were employed, including Chi-square (χ^2) with its associated degrees of freedom (*df*) and significance value (*p*). Additionally, other model fit indices were used, such as the comparative fit index (CFI), the Tucker-Lewis index (TLI), root mean square error of approximation (RMSEA), and standardized root mean square residual (SRMR). Research suggests the following values for acceptable model fit: $X^2/df < 3$ (Abbad, 2021), CFI > 0.90, TLI > 0.90 (Byrne, 2010), RMSEA < 0.08 (Schreiber et al., 2006), and SRMR < 0.08 (Hair et al., 2022). The results showed acceptable fit for the Nordic model ($X^2 = 909.212$, $df = 426$, $X^2/df = 2.134$, $p < 0.001$; CFI = 0.920; TLI = 0.907; RMSEA = 0.062 [90% CI = 0.056, 0.068]; SRMR = 0.059) and the USA model ($X^2 = 744.761$, $df = 426$, $X^2/df = 1.748$, $p < 0.001$; CFI = 0.955; TLI = 0.947; RMSEA = 0.051 [90% CI = 0.045, 0.057]; SRMR = 0.048). These results establish a basis for further analysis of the structural relationships between the constructs.

4.4.3 Path analysis and hypotheses testing

The evaluation of the structural model involved examining all hypothesized relationships. This assessment focused on the strength and direction of the path coefficient values (β), followed by an analysis of *t*- and *p*-values to determine the significance of these relationships. The statistical significance of the path coefficients was determined using a bootstrapping procedure with 5000 iterations. It should be noted that this study adhered to the guidelines by Hair et al. (2022) that consider a path coefficient different from zero at a 5% significance level (α = 0.05; two-tailed test) when the resulting empirical *t*-value exceeds ± 1.96. The resulting *t*-values were subsequently used to evaluate the proposed hypotheses (see Table 9 and Fig. 2).

Three constructs emerged as predictors when evaluating the determinants of behavioral intention to use ChatGPT in the Nordic sample. The results follow Cohen's (1988) thresholds for effect sizes (f^2) of 0.02, 0.15, and 0.35 for small, medium, and large effects, respectively. Performance expectancy emerged as the strongest predictor of behavioral intention (β = 0.379, $p < 0.01$, $t = 7.377$; $f^2 = 0.219$). Furthermore, both habit (β = 0.361, $p < 0.01$, $t = 7.292$; $f^2 = 0.203$) and hedonic motivation (β = 0.121, $p < 0.01$, $t = 2.919$; $f^2 = 0.033$) were predictors of behavioral intention, with habit demonstrating a notably higher effect. The strongest predictor of ChatGPT use was behavioral intention (β = 0.385, $p < 0.01$, $t = 6.429$; $f^2 = 0.143$), followed by habit (β = 0.357, $p < 0.01$, $t = 6.190$; $f^2 = 0.130$). Personal innovativeness showed a tendency to positively influence use behavior (β = 0.086, $p = 0.060$, $t = 1.878$; $f^2 = 0.013$).

Analysis of the USA sample revealed parallels and divergences from the Nordic results. Consistent with the Nordic sample, performance expectancy was the most

Table 9 Path coefficient, effect size, and significance for the construct relationships in the (a) Nordic and (b) USA models

H	Path	Coefficient (β)	Effect size (f^2)	*t*-value	*p*-value	Supported
(a) Nordic						
H1	PE → BI	0.379	0.219	7.377	0.000	Yes
H2	EE → BI	0.036	0.002	0.754	0.451	No
H3	SI → BI	0.071	0.011	1.794	0.073	No
H4	FC → BI	0.063	0.008	1.400	0.162	No
H5	FC → UB	0.017	0.001	0.371	0.711	No
H6	HM → BI	0.121	0.033	2.919	0.004	Yes
H7	PV → BI	0.052	0.006	1.272	0.203	No
H8	HB → BI	0.361	0.203	7.292	0.000	Yes
H9	HB → UB	0.357	0.130	6.190	0.000	Yes
H10	PI → BI	0.012	0.000	0.265	0.791	No
H11	PI → UB	0.086	0.013	1.878	0.060	No
H12	BI → UB	0.385	0.143	6.429	0.000	Yes
(b) USA						
H1	PE → BI	0.387	0.283	8.103	0.000	Yes
H2	EE → BI	0.014	0.000	0.278	0.781	No
H3	SI → BI	0.097	0.022	2.284	0.022	Yes
H4	FC → BI	0.072	0.010	1.874	0.061	No
H5	FC → UB	0.072	0.007	1.158	0.117	No
H6	HM → BI	0.038	0.003	0.762	0.446	No
H7	PV → BI	0.037	0.004	1.008	0.313	No
H8	HB → BI	0.315	0.180	6.332	0.000	Yes
H9	HB → UB	0.325	0.090	4.846	0.000	Yes

Table 9 (continued)

H	Path	Coefficient (β)	Effect size (f^2)	*t*-value	*p*-value	Supported
H10	PI → BI	0.144	0.048	3.438	0.001	Yes
H11	PI → UB	0.023	0.001	0.402	0.688	No
H12	BI → UB	0.345	0.080	4.430	0.000	Yes

PE, Performance expectancy; *EE*, Effort expectancy; *SI*, Social influence; *FC*, Facilitating conditions; *HM*, Hedonic motivation; *PV*, Price value; *HB*, Habit; *PI*, Personal innovativeness; *BI*, Behavioral intention; *UB*, Use behavior

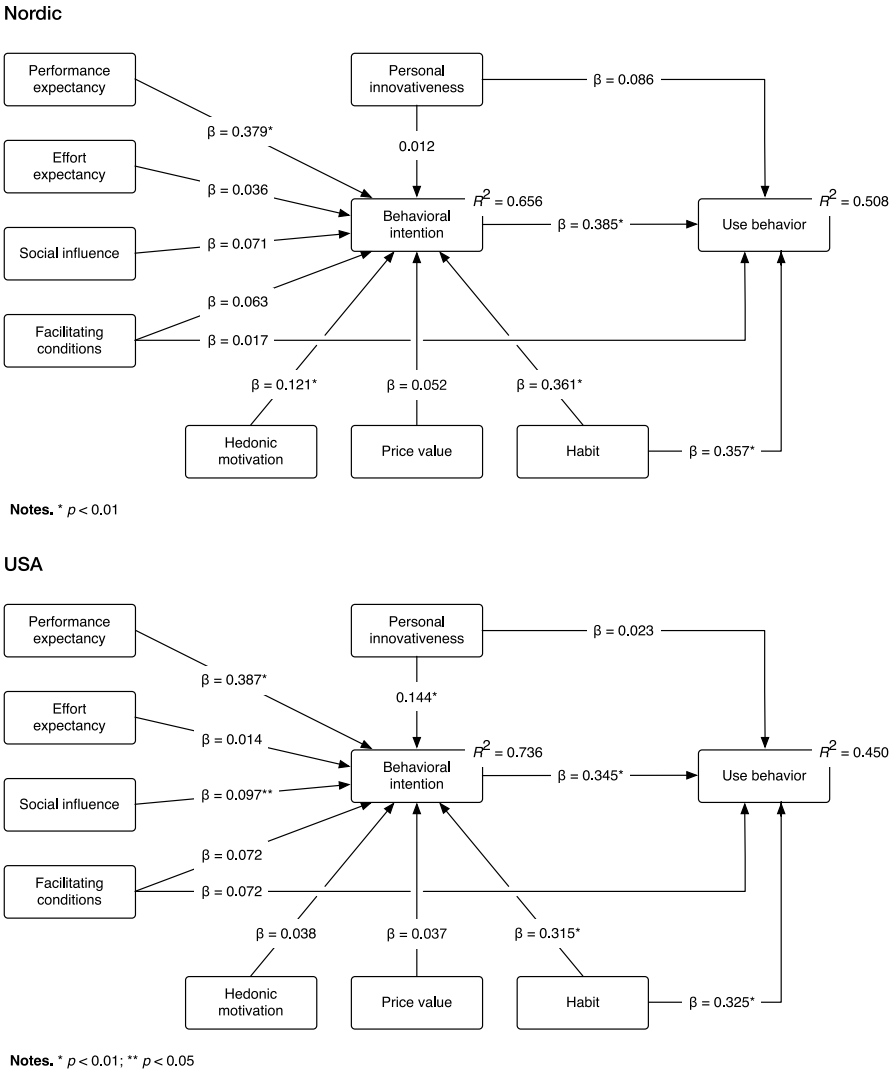

Fig. 2 Hypotheses testing and findings of the structural model for the Nordic countries and the USA

robust predictor of behavioral intention (β = 0.387, $p < 0.01$, $t = 8.103$; $f^2 = 0.283$), followed by habit (β = 0.315, $p < 0.01$, $t = 6.332$; $f^2 = 0.180$). Unlike in the Nordic sample, personal innovativeness was a predictor of behavioral intention in the USA sample (β = 0.144, $p < 0.01$, $t = 3.438$; $f^2 = 0.0048$). Further examining the predictors of behavioral intention, the USA sample revealed positive relationships for social influence (β = 0.097, $p = 0.011$, $t = 2.284$; $f^2 = 0.022$). Facilitating conditions showed a tendency to positively influence behavioral intention (β = 0.072, $p = 0.061$, $t = 1.874$; $f^2 = 0.010$). These relationships were less evident in the Nordic context, where both variables showed weaker associations with behavioral

intention. Finally, the findings revealed that both habit ($\beta = 0.325$, $p < 0.01$, $t = 4.846$; $f^2 = 0.090$) and behavioral intention ($\beta = 0.345$, $p < 0.01$, $t = 4.430$; $f^2 = 0.080$) showed substantial positive relationships with ChatGPT usage.

4.4.4 Model quality and predictive relevance

The quality of the Nordic and USA models was evaluated using in-sample explanatory power (R^2), out-of-sample predictive power ($PLS_{predict}$), and the cross-validated predictive ability test (CVPAT). The explanatory power of the Nordic and USA models was assessed using the coefficient of determination (R^2). This metric quantifies the proportion of variance in the endogenous variables accounted for by the exogenous variables. The scale of R^2 ranged from 0 to 1, with higher values denoting greater explanatory power. In the Nordic model, all factors (performance expectancy, effort expectancy, social influence, facilitating conditions, hedonic motivation, price value, habit, personal innovativeness) accounted for 65.6% of the variance in behavioral intention, while a subset of variables (facilitating conditions, personal innovativeness, habit, behavioral intention) explained 50.8% of the variance in the actual use of ChatGPT in higher education. The USA model showed different results, with all factors explaining 73.6% of the variance in behavioral intention and the subset of variables accounting for 45.0% of the variance in the actual use of ChatGPT (see Table 10).

Additionally, the $PLS_{predict}$ procedure was employed to calculate $Q^2_{predict}$ to assess the predictive relevance of endogenous constructs. The analysis was conducted using tenfold cross-validation with 10 repetitions and a fixed seed for the random number generator. Because the prediction error distribution was symmetric, root mean square error (RMSE) was used as the evaluation measure (Shmueli et al., 2019). Mean absolute error (MAE) is included as a reference. The analysis revealed predictive relevance for both the Nordic and USA models. In the Nordic model, behavioral intention and use behavior had values above zero, indicating predictive relevance. Similarly, the USA model demonstrated predictive relevance for behavioral intention and use behavior (see Table 10).

To conclude the assessment of predictive power, CVPAT was calculated for both behavioral intention and use behavior in the Nordic and USA models (Liengaard et al., 2021; Sharma et al., 2023). The results demonstrated that all PLS-SEM predictions outperformed the naïve indicator average (IA) for both models. However, the predictive power was not strong, as indicated by the linear model (LM)

Table 10 Explanatory power and predictive relevance of the endogenous constructs in the Nordic and USA models

Model	Construct	R^2	Adjusted R^2	$Q^2_{predict}$	RMSE	MAE
Nordic	BI	0.656	0.647	0.629	0.614	0.484
	UB	0.508	0.502	0.440	0.752	0.608
USA	BI	0.736	0.729	0.718	0.536	0.425
	UB	0.450	0.442	0.395	0.783	0.636

BI, Behavioral intention; *UB*, Use behavior; *RMSE*, Root mean square error; *MAE*, Mean absolute error

results. Specifically, in the Nordic model, behavioral intention (IA = −1.041, p_{IA} = 0.000, LM = −0.006, p_{LM} = 0.864) and use behavior (IA = −1.333, p_{IA} = 0.000, LM = −0.036, p_{LM} = 0.658) showed clear improvement over the IA benchmark but were less conclusive given the LM results. Similarly, in the USA model, behavioral intention (IA = −1.594, p_{IA} = 0.000, LM = −0.031, p_{LM} = 0.419) and use behavior (IA = −1.301, p_{IA} = 0.000, LM = −0.133, p_{LM} = 0.65) demonstrated the same pattern.

4.4.5 Importance-performance map analysis (IPMA)

Importance-performance map analysis aims to extend the results of PLS-SEM by identifying areas for improvement. This analysis is conducted by comparing the relative importance of the constructs with their performance. Importance is represented by the path coefficients, with higher values indicating greater importance. Performance is quantified on a scale of 0 to 100, with higher scores reflecting better performance. Table 11 presents the results of the analysis.

The Nordic results indicated that performance expectancy holds the highest importance on behavioral intention to use ChatGPT (β = 0.379), while personal innovativeness exhibited the lowest importance (β = 0.012). Interestingly, two constructs (behavioral intention and habit) demonstrated good importance regarding the use behavior of ChatGPT, with habit (β = 0.496) showing higher importance than behavioral intention (β = 0.385). The findings for the USA showed that the construct with the highest importance for behavioral intention was performance expectancy (β = 0.387), mirroring the Nordic results, while the smallest importance was attributed to effort expectancy (β = 0.014). Concerning the use behavior of ChatGPT in the USA, the pattern was the same and habit (β = 0.325) outweighed behavioral intention (β = 0.345) in importance. Regarding performance metrics, the Nordic

Table 11 Importance-performance map analysis of behavioral intention and use behavior of ChatGPT

Nordic				USA			
Construct	Performance	Importance		Construct	Performance	Importance	
		BI	UB			BI	UB
EE	74.731	0.036		EE	78.289	0.014	
FC	74.115	0.063	0.007	FC	75.219	0.072	0.097
PE	70.630	0.379		PI	71.336	0.144	0.073
PI	69.399	0.012	0.082	HM	69.262	0.038	
HM	67.455	0.121		PV	69.123	0.037	
PV	64.276	0.052		PE	67.740	0.387	
BI	64.132		0.385	BI	63.057		0.345
SI	43.731	0.071		SI	47.706	0.097	
UB	41.610			UB	47.414		
HB	36.938	0.361	0.496	HB	37.363	0.315	0.434

PE, Performance expectancy; *EE*, Effort expectancy; *SI*, Social influence; *FC*, Facilitating conditions; *HM*, Hedonic motivation; *PV*, Price value; *HB*, Habit; *PI*, Personal innovativeness; *BI*, Behavioral intention; *UB*, Use behavior

findings revealed effort expectancy as the top performer with a score of 74.731, while habit had the lowest score of 36.938. Similarly, the USA data showed effort expectancy leading with 78.289 and habit was found to be the lowest with a score of 37.363.

5 Discussion

5.1 Theoretical contributions

This study explored how factors in an extended UTAUT model influenced higher education students' behavioral intention and use of ChatGPT across the Nordic countries and the USA. The intercontinental perspective in this study extends previous research on ChatGPT, which has primarily focused on single-country contexts (e.g., Biloš & Budimir, 2024; Foroughi et al., 2023; Habibi et al., 2023; Polyportis & Pahos, 2024; Strzelecki, 2023).

Concerning the first research question, the findings revealed a set of predictors influencing higher education students' behavioral intention and use behavior of ChatGPT. In the Nordic model, performance expectancy, hedonic motivation, and habit were predictors of behavioral intention to use ChatGPT. The USA model revealed four predictors of behavioral intention: performance expectancy, social influence, habit, and personal innovativeness. For both regions, behavioral intention and habit were the only predictors of use behavior. Regarding the second research question, there were differences in the factors driving behavioral intention. In the Nordic model, hedonic motivation emerged as a significant predictor of behavioral intention, while this factor was not influential in the USA. Conversely, the USA model showed that social influence and personal innovativeness had an impact on behavioral intention, factors that did not play a significant role in the Nordic countries. The strengths of the common predictors also varied between the regions. For instance, performance expectancy had a slightly more substantial effect in the USA than in the Nordic countries. Additionally, the model explained more variance in behavioral intention for the USA than for the Nordic countries, while the opposite was true for use behavior.

Performance expectancy was the most robust predictor of students' behavioral intention across both regions. This finding supports hypothesis H1 and aligns with prior studies across various AI applications in education (An, Chai, Li, Zhou, & Yang, 2023; Cortez et al., 2024; Dahri et al., 2024). Students appeared to recognize the benefits of ChatGPT in improving their academic performance, such as more efficient learning processes, faster completion of assignments, and improved productivity. Notably, this shared perception among students transcends regional boundaries and underscores a common view of ChatGPT as a catalyst in their academic studies.

In contrast, effort expectancy did not emerge as a predictor of behavioral intention, leading to the rejection of hypothesis H2. This finding corroborates and challenges the existing research. This result aligns with certain studies on ChatGPT (Habibi et al., 2023), but is in contrast to other studies examining both ChatGPT

specifically (Strzelecki & ElArabawy, 2024) and AI technologies more broadly (Kwak et al., 2022; Tran et al., 2021; Wu et al., 2022). The weak relationship observed may be rooted in the demographic composition of the two samples, mainly consisting of Millennials and Generation Z (Dimock, 2019). Having grown up in an era of digital ubiquity, these groups likely perceive online services as being naturally integrated into their lives. Consequently, students may perceive the use of ChatGPT in academic contexts as requiring little effort to adopt in their study practices.

The findings further revealed that social influence predicted behavioral intention to use ChatGPT in the USA, but not in the Nordic countries. However, the small difference in path coefficients indicates that this interpretation is speculative. Further research is therefore needed to better understand the role of social influence in ChatGPT adoption across different regional contexts. This evidence partially supports hypothesis H3 and provides evidence for both sides of the ongoing debate on the role of social influence in educational technology adoption. This result aligns with studies showing that social influence affects AI adoption in education (Strzelecki, 2023; Tran et al., 2021; Wu et al., 2022), while also supporting research findings that it is less influential for educational technologies such as m-learning (Venkataraman & Ramasamy, 2018) and interactive whiteboards (Wong et al., 2013). In the USA, the role of social influence in behavioral intention to use ChatGPT could be explained by the competitive and socially interconnected nature of U.S. higher education (Bound et al., 2009; Lorbeer, 2020). This environment may foster a climate in which students are likely to be influenced by peers' technology choices to maintain a competitive edge in academic performance. Conversely, the weak relationship between social influence and behavioral intention in Nordic countries can be attributed to the individualistic and autonomous learning culture prevalent in their educational systems (Klette, 2018; Minkov & Kaasa, 2022). In these environments, students are often encouraged to make independent decisions regarding study methods and digital technologies.

Facilitating conditions showed a tendency to positively influence behavioral intention in the USA, but showed no impact in the Nordic countries. While this result does not support hypothesis H4, it partially corroborates studies showing the importance of facilitating conditions in AI adoption for educational purposes (Foroughi et al., 2023; Habibi et al., 2023; Kwak et al., 2022). Concurrently, this also supports research that found facilitating conditions less influential on the intention to use AI (An, Chai, Li, Zhou, & Yang, 2023; Weiqi et al., 2024). No support was found for hypothesis H5 concerning facilitating conditions and use behavior. The observed relationship between facilitating conditions and behavioral intention in this research is similar to that of a study involving Qatar and the USA (El-Masri & Tarhini, 2017). That study found a stronger association between facilitating conditions and behavioral intention for American students than for their Qatari counterparts (El-Masri & Tarhini, 2017). Although both the Nordic countries and Qatar are developing nations (Mohamed et al., 2022), American students appear to be more influenced by the availability of resources, technical support, and compatibility with existing technologies when considering the use of ChatGPT. Further investigation is required to explore the underlying reasons for these differences, including variations in AI literacy, educational policies, and technological infrastructure.

Unlike in the American context, hedonic motivation predicted the behavioral intention to use ChatGPT in the Nordic countries. This finding partially supports hypothesis H6 and aligns with studies that emphasize the role of hedonic motivation in AI adoption within educational contexts (Cai et al., 2023; Li & Qin, 2023; Vitezić & Perić, 2021). The discrepancy between these regions may stem from differing educational structures. Students in the Nordic countries usually study one or two courses concurrently, totaling four courses per semester, with few comprehensive assignments. Importantly, higher education in Nordic countries is typically free, and students can receive non-repayable financial aid and repayable loans from their governments to support their living expenses while studying (Thomsen et al., 2017). This combination of a less intensive course structure, free education, and financial support can potentially free time to explore the hedonic aspects of ChatGPT. By contrast, American students often take four or more courses simultaneously with weekly assignments, midterms, and finals (Huntington-Klein & Gill, 2021). Moreover, many American students need to pay for their education and work to cover tuition and living expenses (Brint, 2022). This heavier workload, financial pressure, and cost of education may lead American students to view ChatGPT primarily as a utilitarian service rather than a source of enjoyment.

Price value did not influence the behavioral intention to use ChatGPT among students in the Nordic countries and the USA. As a result, hypothesis H7 was rejected for both regions. Contrary to some studies on AI adoption (Wang & Zhang, 2023; Zhu et al., 2024), the findings suggest that students do not perceive ChatGPT as offering sufficient value relative to its cost. A primary factor behind this finding is likely the availability of a free version of ChatGPT, which offers text-based interactions sufficient for most students' academic needs. The paid version of the AI service, priced at $20 per month, offers additional input capabilities, such as audio and visual elements, as well as increased message limits. However, these features do not appear to provide sufficient perceived value to justify the monthly cost for most students. While a $20 monthly fee might be manageable occasionally, the cumulative annual cost of $240 could be considered unnecessary for many students (Hirabayashi et al., 2024).

In support of hypothesis H8, habit showed a positive relationship with students' behavioral intention to use ChatGPT in both regions. This finding aligns with research highlighting the role of habitual behaviors in the adoption of AI in academic environments (Almahri et al., 2024; Lavidas et al., 2024; Salifu et al., 2024; Sebastián et al., 2022). The influence of habit may be attributed to the widespread availability of AI services and the perceived academic value of ChatGPT among students to aid them in their learning processes and completing various academic tasks. The impact of habit also influenced use behavior. This observed relationship supported hypothesis H9 and was consistent with research on the actual use of ChatGPT (Lavidas et al., 2024; Salifu et al., 2024). The automatic nature of habitual behavior is often triggered by specific contexts or cues. Certain academic tasks or challenges may prompt students to employ ChatGPT. The common patterns in AI adoption habits across the Nordic countries and the USA can be attributed to several interrelated factors. First, students in these regions are predominantly digital natives who have become immersed in digital technology. Second, they take advantage of

their free access to ChatGPT, which has become increasingly integrated into the academic environment. Finally, AI services empower students with previously unattainable problem-solving capabilities (e.g., generating ideas and summarizing articles) that aid them in addressing academic challenges with greater autonomy.

The analysis of personal innovativeness revealed varied relationships across regions. Hypothesis H10, which predicted an impact on behavioral intention, was partially supported. Hypothesis H11, which predicted an impact on use behavior, was not supported. Contrary to expectations, personal innovativeness did not influence behavioral intention or use behavior in the Nordic countries. In contrast, personal innovativeness predicted behavioral intention for the USA, although its impact on use behavior remained weak. The results in the USA align with studies that emphasize the role of personal innovativeness in ChatGPT adoption (Arif et al., 2024; Biloš & Budimir, 2024; Jo, 2023; Strzelecki, 2023). The insignificance of personal innovativeness in the Nordic countries also supports other studies on ChatGPT (Foroughi et al., 2023; Strzelecki, 2024). This divergence in behavioral intention may be attributed to individual motivation. Research has suggested that American students, from upper secondary school through higher education, typically display higher levels of individual motivation than their Nordic and international counterparts (Karaman & Watson, 2017; OECD, 2017). Given that higher individual motivation can promote innovation performance (Jiang et al., 2023), this could drive both personal innovativeness and the behavioral intention to use ChatGPT among American students. The discrepancy between personal innovativeness predicting behavioral intention, and not use behavior in the USA, suggests that students with high personal innovativeness may be more inclined to consider using ChatGPT. However, other factors, such as actual need, peer and faculty influence, or fear of over-reliance, which could inhibit skill development (e.g., critical thinking, writing, and problem-solving), may determine actual usage. The relationships between individual motivation, personal innovativeness, intention to use ChatGPT, and actual use behavior, as well as the factors influencing these connections, require further investigation to understand their interplay.

Lastly, the results confirmed hypothesis H12 and revealed that behavioral intention predicted use behavior in the Nordic countries and the USA. This observation validates the fundamental principles of the UTAUT2 framework (Venkatesh et al., 2012) and underscores the stability of the relationship between these two constructs across regions and academic settings. These consistent outcomes corroborate research examining the influence of behavioral intention on the actual use of ChatGPT (Acosta-Enriquez et al., 2024; Gulati et al., 2024; Jo, 2023; Salifu et al., 2024).

5.2 Practical implications

This study carries several implications for educators and policymakers in higher education. For educators, the findings can be used to understand students' behavior, make informed decisions about curriculum design, and integrate AI chatbots into courses. Given that performance expectancy was the strongest predictor

of behavioral intention to use ChatGPT in both regions, educators could focus on demonstrating the benefits of AI services to students through hands-on workshops. This necessitates educators to design course materials and assignments that showcase the practical potential of these services. For example, in a web development course, educators might showcase how ChatGPT can assist with debugging code and explaining complex programming concepts. Educators can also assign students to use ChatGPT to brainstorm research questions and outline a thesis structure when teaching a methodology course.

While integrating such practical applications, educators should carefully consider the potential consequences. The utilitarian view of ChatGPT among American students may be driven by practical constraints and raises critical questions about students' learning and creativity. Recent policy analysis by McDonald et al. (2025) found that 63% of US universities actively encourage ChatGPT use. In many of these institutions, students are required to provide prompt logs and process descriptions to document their use. Educators could integrate documentation requirements to enhance learning outcomes while promoting critical engagement with AI services. These documentation requirements reflect a broader concern that students might become overly reliant on AI services at the expense of developing traditional learning skills and creative thinking. A potential downside of ChatGPT is that it could take over the role of thinking for students, turning them into passive consumers rather than active learners who develop the ability to think critically and solve problems. This is particularly pertinent in contexts where educational and financial pressures may lead to the predominantly instrumental use of AI services. To mitigate these risks, educators can design assignments that require students to solve these independently before using AI and have them document their reasoning process both when working independently and when using AI services.

The findings may offer insights for policymakers considering AI integration in higher education. The varying results of facilitating conditions between students in the Nordic countries and the USA suggest the importance of understanding regional differences when developing educational approaches. Given that the relationship between facilitating conditions and behavioral intention to use ChatGPT was weaker for Nordic students, policymakers in this region may need to consider initiatives. These could include improving the current IT infrastructure, addressing any underlying skepticism or resistance to AI adoption among students and educators, offering ongoing faculty development programs to ensure that educators can guide students to use AI chatbots responsibly, and integrating AI literacy into existing curricula. Such integration needs to be balanced against potential legal constraints, particularly in the Nordic countries. These include European Union (EU) regulations, such as the General Data Protection Regulation (GDPR) and the EU AI Act.

Integrating AI services with existing learning management systems (LMSs) may present challenges. Many LMSs are proprietary, secured through national procurement agreements, and deeply embedded in institutional infrastructure. Consequently, they cannot be easily replaced or modified to integrate emerging technologies. Navigating these issues may require policymakers to adopt a holistic strategy that balances technological integration with legal compliance, institutional capabilities,

educational goals, and potentially a high cost of an enterprise subscription that would provide students and educators access to premium versions of AI services.

5.3 Methodological considerations

Several methodological considerations should be noted when interpreting the results. First, using Prolific for participant recruitment may have introduced sample bias. While the platform offers advantages regarding participants' diversity and data quality, its user base may not fully represent the broader student population in the Nordic countries and the USA. Specifically, users of such platforms may possess the characteristics that influence their technology adoption. Second, the uneven distribution of participants across the Nordic countries, with a notably small sample from Iceland, may have skewed the results for the Nordic region. This imbalance could have overrepresented the perspectives of students in countries with more participants, such as Sweden. Third, this study provides a snapshot of ChatGPT adoption at a single point in time. While this approach offers valuable insights, a longitudinal study could provide a comprehensive understanding of how students' use of ChatGPT evolves throughout their academic journey.

5.4 Limitations and future research

Despite the rigorous nature of the research design and quantitative methodology, some limitations come to the fore. First, while this study controlled for several constructs, there may be unaccounted factors influencing ChatGPT adoption, such as the field of study and prior experiences with AI services. Future work could address this limitation by collecting data on participants' academic majors and conducting assessments of their AI familiarity.

Second, the UTAUT2 framework offered insights into technology adoption factors, but mainly focused on positive behavior aspects. This limited the current study's coverage of the potential concerns or negative implications associated with ChatGPT use in higher education. Although this study gained quantitative insights into adoption patterns, it did not capture qualitative perspectives on the benefits or drawbacks of ChatGPT use. Future research could use open-ended questions or interviews as a complement to explore both the positive and negative aspects of ChatGPT use in higher education.

Third, the focus on ChatGPT as a single AI service may have limited the ecological validity of this study. In real-world educational settings, students often use multiple AI chatbots, such as Gemini, Claude, and Copilot. Future directions could strengthen the ecological validity by examining and comparing the adoption and use patterns of several AI chatbots. Fourth, while this study focused on direct effects, examining indirect effects through behavioral intention could reveal how factors such as performance expectancy and social influence indirectly impact use behavior. Additionally, investigating whether behavioral intention moderates the relationships between other constructs could reveal more complex interaction patterns in

technology adoption. Finally, this study focused on higher education students and future work could apply the proposed model to other educational contexts, such as K-12 education.

6 Conclusions

The rapid evolution of AI services presents opportunities and challenges for educational institutions. While ChatGPT offers promising advantages for learning, critical questions remain about its long-term effects on education. In general, as AI services advance, higher education needs to continuously evaluate their impact on learning and adapt policies, curricula, and pedagogical practices accordingly. This study underscores the need for educators to consider global trends and local educational contexts when developing policies and practices for AI adoption in academic settings. While universities can prepare students for a future where AI is integral to both education and professional life, they must also carefully monitor and assess its impact on fundamental learning processes. AI services raise important questions about how they affect students' development of critical thinking skills, capacity for independent analysis, and ability to engage in deep learning. AI services should be thoughtfully designed into educational approaches to support students' active learning without creating dependency. A balanced approach can support students in developing their practical skills and fostering lifelong learning mindsets.

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Data Availability The participants of this study did not give written consent for their data to be shared publicly, so due to the sensitive nature of the research supporting data is not available.

Declarations

Competing Interests The authors confirm that there are no relevant financial or non-financial competing interests to report.

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Authors and Affiliations

Montathar Faraon[1] · Kari Rönkkö[1] · Marcelo Milrad[2] · Eric Tsui[3]

✉ Montathar Faraon
montathar.faraon@hkr.se

Kari Rönkkö
kari.ronkko@hkr.se

Marcelo Milrad
marcelo.milrad@lnu.se

Eric Tsui
eric.tsui@polyu.edu.hk

[1] Department of Design, Kristianstad University, Kristianstad 291 88, Sweden

[2] Department of Computer Science and Media Technology, Linnaeus University, 352 52, Vaxjo, Sweden

[3] Educational Research Centre, The Hong Kong Polytechnic University, Hung Hom, Kowloon, Hong Kong